Inter-IIT Tech Meet 15.0

Quantitative Trading - Track A

Volatility-Adjusted Cross-Sectional Reversal Strategy

Strategy and Backtest Report

Submitted by: Kashishdeep Kaur
Identifier: 142401020

Track: A - Cross-Sectional Trading
Initial Capital: $100,000
Evaluation Dataset: Dataset_PS-A.csv

This report describes the strategy implementation and its historical backtest results. The reported performance is historical and in-sample; it should not be interpreted as a guarantee of future performance.

Contents

1 Executive Summary **2**

2 Strategy Overview **2**

2.1 Motivation 2

2.2 Universe and Portfolio Construction 2

2.3 Signal Definition 3

2.4 Volatility-Adjusted Weighting 3

3 Implementation and Execution **3**

4 Backtest Results **4**

4.1 Portfolio Performance 4

4.2 Trading Activity and Costs 4

5 Risk Controls **4**

6 Limitations and Validation **5**

7 Conclusion **6**

1 Executive Summary

This report presents a long-only, cross-sectional reversal strategy for Track A of the Inter-IIT Tech Meet 15.0 Quantitative Trading selection. The strategy identifies assets with relatively weak recent returns and allocates capital to a small number of these assets, adjusting position weights for recent volatility.

The portfolio selects five assets at each rebalance. Position weights depend on both the relative strength of the reversal signal and the inverse of the asset's estimated volatility. The resulting weights are normalized to keep gross portfolio exposure at or below 100% of portfolio equity.

The final recorded historical backtest used an initial capital of $100,000 and reported final equity of $585,093.28. This corresponds to a total return of 485.09%, a compound annual growth rate (CAGR) of 34.98%, and a Sharpe ratio of 1.40. The maximum reported drawdown was 32.85%.

These results are encouraging as an initial historical evaluation, but they do not establish that the strategy will generalize to unseen data. In particular, the evaluation uses zero execution slippage, and no walk-forward validation or fully untouched holdout evaluation has been completed for the results reported here.

2 Strategy Overview

2.1 Motivation

Cross-sectional reversal is based on the hypothesis that assets with relatively poor recent performance may subsequently recover relative to other assets. This effect is not guaranteed to persist and can vary across market regimes.

The strategy ranks assets using recent returns and takes long positions in a small set of assets with the weakest recent performance. Volatility adjustment is used to reduce the relative weight of assets with higher estimated price variability.

The strategy is long-only: it does not open short positions.

2.2 Universe and Portfolio Construction

The Track A dataset used for the reported run contained 24 assets across 1,484 trading days. The portfolio construction process is:

1. Estimate each asset's recent return using a 20-step lookback window.
2. Rank assets according to their recent returns, with relatively poor-performing assets receiving stronger reversal signals.
3. Select five assets from the weak-performing end of the cross-sectional ranking.
4. Estimate recent volatility using a 20-step window.
5. Allocate weights using reversal rank strength adjusted by inverse volatility.
6. Normalize the resulting long weights so that total gross exposure does not exceed 1.0.

The implementation uses a 20-step rebalance cadence in `src/engine.py`. This is the cadence used by the submitted strategy implementation; it is not inferred from the original configuration value.

2.3 Signal Definition

Let $P_{i,t}$ denote the price used by the implementation for asset i at time t. The simple return over a lookback window of $L = 20$ steps can be represented as

$$R_{i,t}^{(L)} = \frac{P_{i,t}}{P_{i,t-L}} - 1.$$

The reversal signal is the negative of this recent return:

$$s_{i,t} = -R_{i,t}^{(L)}.$$

A more negative recent return therefore produces a stronger positive reversal preference. The actual ranking and signal calculation are implemented in `src/engine.py`.

2.4 Volatility-Adjusted Weighting

Let $\sigma_{i,t}$ denote the recent volatility estimate for asset i, and let $q_{i,t}$ represent its positive rank-strength weight among the selected assets. The raw allocation score is

$$\widetilde{w}_{i,t} = \frac{q_{i,t}}{\sigma_{i,t}},$$

for selected assets with valid positive volatility estimates.

The scores are normalized across the selected long positions:

$$w_{i,t} = \frac{\widetilde{w}_{i,t}}{\sum_{j \in \mathcal{L}_t} \widetilde{w}_{j,t}},$$

where $\mathcal{L}_t$ is the set of selected assets. This expression describes the intended weighting principle; implementation-specific handling of invalid or zero volatility estimates is governed by the strategy code.

The resulting weights are long-only and normalized to satisfy

$$w_{i,t} \geq 0, \qquad \sum_i |w_{i,t}| \leq 1.$$

This constraint limits gross exposure but does not eliminate market risk or guarantee low portfolio volatility.

3 Implementation and Execution

The strategy is implemented through the required `ParticipantStrategy` interface in `src/engine.py`. The backtest was run using the provided Track A backtesting framework and the dataset `data/Dataset_PS-A.csv`.

The reported run processed 1,484 trading steps and 24 assets. The framework's documented execution convention fills signals generated at step t at the Open of step $t + 1$. This convention is intended to prevent execution at the same bar's opening price based on a signal that is only available later in that bar.

The strategy uses the following settings for the reported run:

- **Return lookback:** 20 steps.
- **Volatility lookback:** 20 steps.

- **Selected positions:** Five long positions.
- **Rebalance cadence:** 20 steps, as hardcoded in the strategy engine.
- **Commission:** 0.0001 per traded value, equivalent to 1 basis point.
- **Execution slippage:** 0.0000 in the configuration.
- **Initial capital:** $100,000.

The zero-slippage assumption is optimistic compared with real execution. Consequently, the reported results may overstate achievable performance after market impact, bid–ask spreads, liquidity constraints, and other execution frictions.

4 Backtest Results

4.1 Portfolio Performance

Table 1 summarizes the metrics reported by the final verification run.

Table 1: Historical portfolio performance

Metric	Reported value
Initial capital	$100,000.00
Final equity	$585,093.28
Total return	485.09%
CAGR	34.98%
Sharpe ratio	1.40
Sharpe 95% confidence interval	[0.67, 2.21]
Sortino ratio	1.82
Maximum drawdown	32.85%
Calmar ratio	1.06
Daily win rate	53.74%

The portfolio grew substantially over the evaluated historical period. However, the maximum drawdown of 32.85% indicates that the return profile includes material downside risk. The return and risk metrics should therefore be considered together rather than judging the strategy on total return alone.

The reported Sharpe confidence interval reflects uncertainty in the Sharpe estimate under the backtester's estimation procedure. It does not, by itself, establish statistically significant alpha or prove that the strategy will perform similarly in the future.

4.2 Trading Activity and Costs

The reported turnover is high relative to the initial capital. This makes transaction-cost assumptions particularly important. Although the run deducts commissions, it assumes no slippage. Actual execution costs could materially reduce performance, especially if the assets are less liquid or trades are executed at unfavorable prices.

5 Risk Controls

The implementation was evaluated against the framework's gross-leverage and non-negative-capital constraints.

Table 2: Trading activity and transaction costs

Metric	Reported value
Total trades	7,099
Buy trades	3,651
Sell trades	3,448
Trading frequency	4.78 trades per day
Market exposure time	98.58%
Total turnover	38,317.15%
Total turnover value	$38,317,149.26
Average trade value	$5,397.54
Commission paid	$3,831.71
Slippage paid	$0.00

Table 3: Reported risk and constraint checks

Metric	Reported value
Mean gross leverage	0.98
Maximum gross leverage	1.00
Gross-leverage violations	0
Account bankruptcy events	0

The backtest reports no gross-leverage violations and no bankruptcy events. These results support compliance with the tested constraints for this run. They do not establish protection against all forms of market risk, execution risk, or extreme price movements.

The strategy remains exposed to broad market movements because it is long-only and does not explicitly neutralize market beta or sector exposures.

6 Limitations and Validation

The results should be interpreted in light of the following limitations.

1. **In-sample evaluation.** The reported performance is based on the historical dataset used during strategy development and evaluation. It is not evidence of out-of-sample profitability.
2. **No completed walk-forward validation.** The reported results do not include a fully documented walk-forward evaluation in which parameters are selected using past data and tested on subsequent unseen periods.
3. **No untouched holdout.** No fully untouched holdout result is reported here. The generalization of the strategy to new market periods remains uncertain.
4. **Optimistic execution assumptions.** Commission is included, but slippage is set to zero. The reported performance may therefore be optimistic.
5. **High turnover.** The large cumulative traded value means that realized transaction costs and liquidity limitations could be important in practice.
6. **Market exposure.** The long-only portfolio may be sensitive to broad market drawdowns. The strategy does not explicitly enforce market neutrality or sector neutrality.

7. **Statistical uncertainty.** Historical performance metrics are estimates and can be affected by market regimes, serial dependence, parameter choices, and the finite length of the dataset.

Further validation should use chronological train/test separation, realistic execution costs, and a frozen strategy evaluated on subsequent data. Any parameter tuning should be confined to the development period rather than repeatedly optimized against the evaluation period.

7 Conclusion

This report presents a long-only, volatility-adjusted cross-sectional reversal strategy implemented for Track A. The strategy selects five assets based on recent relative weakness, adjusts their weights using inverse volatility and rank strength, and normalizes allocations to keep gross exposure at or below one.

The final recorded backtest reports a 485.09% total return, 34.98% CAGR, and 1.40 Sharpe ratio, with a maximum drawdown of 32.85%. The backtester also reports zero gross-leverage violations and no bankruptcy events.

These results provide a historical assessment of the implemented strategy, not proof of future profitability. High turnover, the zero-slippage assumption, the lack of completed walk-forward validation, and the absence of an untouched holdout remain important limitations. Independent out-of-sample evaluation and more realistic transaction-cost modeling are necessary before drawing stronger conclusions about the strategy's robustness.

End of Report